Mathematical Proof I: Deconstructing a Problem[1]

1 The Goal

In this block, our goal is not to learn a new collection of mathematical results. Instead, we will practice how to **approach** a mathematical problem.

When we see a statement such as

$$\text{Assumptions} \quad \Longrightarrow \quad \text{Conclusion},$$

we often want to immediately start manipulating equations.

A better first step is to stop and ask:

1. What exactly am I given?
2. What exactly am I being asked to show?
3. What definitions translate the statement into mathematics?
4. What objects should I introduce?
5. What does each assumption allow me to conclude?

The objective of today's session is to make these questions automatic.

$$\boxed{\text{Statement} \longrightarrow \text{Deconstruction} \longrightarrow \text{Proof Skeleton} \longrightarrow \text{Proof}}$$

A good proof rarely starts with algebra. It usually starts with understanding what the statement actually says.

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2 Step 1: Separate What Is Given from What Must Be Shown

Consider the following statement.

Example 1 (Strict Monotonicity and Injectivity)

Suppose $f : \mathbb{R} \to \mathbb{R}$ is strictly increasing. Show that f is injective.

Before trying to prove the statement, separate it into two parts.

$$\boxed{\text{Given: } f \text{ is strictly increasing}}$$

and

$$\boxed{\text{Goal: } f \text{ is injective.}}$$

Neither statement is yet written in a particularly useful form.

We therefore unpack the relevant definitions.

Since f is strictly increasing,

$$x < y \quad \Longrightarrow \quad f(x) < f(y).$$

To prove that f is injective, we must show

$$f(x) = f(y) \quad \Longrightarrow \quad x = y.$$

The original problem can therefore be rewritten as:

> Take $x, y \in \mathbb{R}$. Suppose $f(x) = f(y)$. Using the fact that
>
> $$x < y \Longrightarrow f(x) < f(y),$$
>
> show that $x = y$.

This version is much closer to a proof.

2.1 What should we introduce?

The goal itself suggests what objects we need.

Since injectivity concerns two arbitrary points x and y, a natural beginning is:

> Take $x, y \in \mathbb{R}$ such that $f(x) = f(y)$.

We can now ask what would happen if $x \neq y$.

If $x \neq y$, then either

$$x < y \qquad \text{or} \qquad y < x.$$

Strict monotonicity would then imply either

$$f(x) < f(y) \qquad \text{or} \qquad f(y) < f(x),$$

both of which contradict $f(x) = f(y)$.

Solution

Take $x, y \in \mathbb{R}$ and suppose $f(x) = f(y)$. Suppose, toward a contradiction, that $x \neq y$. Then either $x < y$ or $y < x$.

If $x < y$, strict monotonicity implies

$$f(x) < f(y),$$

contradicting $f(x) = f(y)$.

Similarly, if $y < x$, then

$$f(y) < f(x),$$

again a contradiction.

Therefore $x = y$, and hence f is injective.

The important part of this example is not the proof itself. It is the deconstruction:

$$\begin{aligned} \text{Given} &: \text{ strictly increasing,} \\ \text{Definition} &: \ x < y \Rightarrow f(x) < f(y), \\ \text{Goal} &: \text{ injective,} \\ \text{Definition} &: \ f(x) = f(y) \Rightarrow x = y. \end{aligned}$$

3 Step 2: Let the Conclusion Tell You How to Start

Very often, the conclusion tells us what the first line of the proof should look like.

For example:

Goal	**Natural first move**
Show $A \subseteq B$	Take arbitrary $x \in A$.
Show $A = B$	Prove $A \subseteq B$ and $B \subseteq A$.
Show f is increasing	Take $x < y$.
Show f is injective	Take x, y such that $f(x) = f(y)$.
Show C is convex	Take $x, y \in C$ and $\lambda \in [0, 1]$.
Show x^* is a maximizer	Take arbitrary feasible x and compare $f(x)$ with $f(x^*)$.

This is not a table to memorize. The general lesson is:

$$\boxed{\text{Unpack the conclusion first.}}$$

Once the conclusion has been translated into its definition, the objects that need to appear in the proof often become obvious.

4 Example: Convex Sets

Convexity will appear repeatedly in first-year microeconomics.

Definition 1 (Convex Set)

A set $C \subseteq \mathbb{R}^n$ is **convex** if for every $x, y \in C$ and every $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \in C.$$

Now consider:

Proposition 1

Let $A, B \subseteq \mathbb{R}^n$ be convex. Then $A \cap B$ is convex.

Do not begin the proof yet.

Instead, deconstruct the statement.

What is given?

$$A \text{ is convex} \qquad \text{and} \qquad B \text{ is convex.}$$

This means that whenever $x, y \in A$,

$$\lambda x + (1-\lambda)y \in A,$$

and whenever $x, y \in B$,

$$\lambda x + (1-\lambda)y \in B.$$

What must be shown?

We must show that $A \cap B$ is convex.

Expanding the definition, we need to show:

$$x, y \in A \cap B, \quad \lambda \in [0,1]$$

implies

$$\lambda x + (1-\lambda)y \in A \cap B.$$

What should the first line be?

The definition tells us directly:

Take arbitrary $x, y \in A \cap B$ and $\lambda \in [0, 1]$.

Now ask what $x, y \in A \cap B$ means.

It means

$$x, y \in A \qquad \text{and} \qquad x, y \in B.$$

We can therefore use the convexity of both sets.

Solution

Take arbitrary $x, y \in A \cap B$ and $\lambda \in [0, 1]$.

Since $x, y \in A \cap B$,

$$x, y \in A \qquad \text{and} \qquad x, y \in B.$$

Since A is convex,

$$\lambda x + (1 - \lambda)y \in A.$$

Since B is convex,

$$\lambda x + (1 - \lambda)y \in B.$$

Therefore,

$$\lambda x + (1 - \lambda)y \in A \cap B.$$

Hence $A \cap B$ is convex.

Notice that essentially every line of the proof came from unpacking a definition.

5 Exercise: Deconstruct Before You Prove

Example 1 (Translation of a Convex Set)

Suppose $C \subseteq \mathbb{R}^n$ is convex and $a \in \mathbb{R}^n$. Define

$$D = \{x + a : x \in C\}.$$

Show that D is convex.

Before writing a proof, answer the following questions.

(a) What exactly is given?

(b) What exactly must be shown?

(c) What objects should you introduce in the first line of the proof?

(d) If $u \in D$, what does the definition of D tell you about u?

(e) If $u, v \in D$, how can you write u and v using points in C?

(f) Where do you expect to use the assumption that C is convex?

6 Step 3: Ask What Each Assumption Gives You

When a theorem contains several assumptions, a useful question is:

$$\boxed{\text{What does each assumption buy me?}}$$

Consider a result that will appear repeatedly in optimization and microeconomic theory.

Proposition 1 (Strict Concavity and Uniqueness)

Let $C \subseteq \mathbb{R}^n$ be convex, and let $f : C \to \mathbb{R}$ be strictly concave. Then f has at most one maximizer on C.

Before proving the result, identify the role played by every piece of the statement.

Assumption 1: C is convex

If $x, y \in C$, then for every $\lambda \in [0, 1]$,

$$\lambda x + (1 - \lambda)y \in C.$$

So convexity tells us that points between feasible alternatives remain feasible.

Assumption 2: f is strictly concave

For distinct $x, y \in C$ and every $\lambda \in (0, 1)$,

$$f\big(\lambda x + (1 - \lambda)y\big) > \lambda f(x) + (1 - \lambda)f(y).$$

So strict concavity gives us a strict inequality.

Goal: At most one maximizer

What does "at most one maximizer" actually mean?

A useful reformulation is:

$$x^*, y^* \in \text{argmax}_{x \in C} f(x) \quad \Longrightarrow \quad x^* = y^*.$$

Alternatively, if we want to reason by contradiction, we could suppose that there exist two distinct maximizers:

$$x^* \neq y^*.$$

If both are maximizers, then

$$f(x^*) = f(y^*) = M$$

for some maximal value M.

The structure of the argument is now almost visible:

$$\begin{gathered} x^*, y^* \text{ are distinct maximizers} \\ \Downarrow \\ \lambda x^* + (1-\lambda) y^* \text{ is feasible} \\ \Downarrow \\ f\big(\lambda x^* + (1-\lambda) y^*\big) > M \\ \Downarrow \\ \text{contradiction.} \end{gathered}$$

Solution

Suppose, toward a contradiction, that $x^* \neq y^*$ are both maximizers of f on C. Then

$$f(x^*) = f(y^*) = M,$$

where M is the maximum value of f on C.

Since C is convex, for any $\lambda \in (0, 1)$,

$$\lambda x^* + (1 - \lambda) y^* \in C.$$

Since f is strictly concave and $x^* \neq y^*$,

$$\begin{aligned} f\big(\lambda x^* + (1-\lambda)y^*\big) &> \lambda f(x^*) + (1-\lambda) f(y^*) \\ &= \lambda M + (1-\lambda) M \\ &= M. \end{aligned}$$

Thus there exists a feasible point whose value is strictly greater than the maximum value M, a contradiction.

Therefore f has at most one maximizer on C.

7 Which Assumption Was Used Where?

The previous proof is useful because each assumption performs a different job.

Information	**What it gives us**
C is convex	The convex combination $$\lambda x^* + (1-\lambda)y^*$$ is feasible.
f is strictly concave	The convex combination has value strictly greater than the weighted average of $f(x^*)$ and $f(y^*)$.
x^*, y^* are maximizers	$$f(x^*) = f(y^*) = M,$$ and no feasible point can have value greater than M.
$x^* \neq y^*$	Allows us to use the *strict* part of strict concavity.

When you are stuck in a proof, it is often useful to ask:

> I have not used assumption X yet. What does assumption X allow me to conclude?

8 Exercise: Positive Definiteness

Positive and negative definite matrices will appear repeatedly in optimization, second-order conditions, quadratic forms, and econometrics.

Definition 1 (Positive Definite Matrix)

A symmetric matrix $A \in \mathbb{R}^{n\times n}$ is **positive definite** if

$$x^\top A x > 0$$

for every $x \neq 0$.

Now consider:

Proposition 1

Suppose A is positive definite. Show that

$$Ax = 0 \quad \Longrightarrow \quad x = 0.$$

Do not prove the result immediately.

First deconstruct it.

(a) What is given about A?

(b) What is given about x?

(c) What exactly must be shown?

(d) If $Ax = 0$, what can you say about

$$x^\top Ax?$$

(e) If $x \neq 0$, what does positive definiteness imply about

$$x^\top Ax?$$

(f) Can both conclusions be true at the same time?

Solution

Suppose $Ax = 0$. Then

$$x^\top Ax = x^\top 0 = 0.$$

Suppose $x \neq 0$. Since A is positive definite,

$$x^\top Ax > 0,$$

which contradicts $x^\top Ax = 0$.

Therefore $x = 0$.

9 Step 4: Translate Complicated Notation One Piece at a Time

First-year economics often introduces familiar ideas using dense notation.

For example, suppose

$$V(p, w) = \max_{x \in B(p,w)} u(x),$$

where

$$B(p, w) = \left\{ x \in \mathbb{R}^L_+ : p \cdot x \leq w \right\}.$$

Before doing any optimization, make sure you can read the statement.

- $B(p, w)$ is a set of feasible choices.
- $x \in B(p, w)$ means that x satisfies the budget constraint.
- $V(p, w)$ is the highest utility attainable among feasible choices.
- If $x^* \in \text{argmax}_{x \in B(p,w)} u(x)$, then x^* is a feasible choice attaining that highest value.

Thus

$$x^* \in \text{argmax}_{x \in B(p,w)} u(x)$$

contains two pieces of information:

$$x^* \in B(p,w)$$

and

$$u(x^*) \geq u(x) \qquad \forall x \in B(p,w).$$

This translation is often more important than any subsequent calculation.

10 Exercise: Read an Optimization Statement

Example 1 (A Choice Problem)

Let X be a set. For each $s \in X$, let $B(s) \subseteq \mathbb{R}^n$, and define

$$V(s) = \max_{a \in B(s)} u(a).$$

Suppose

$$a^* \in \text{argmax}_{a \in B(s)} u(a).$$

Without doing any optimization, explain precisely what each statement means.

(a) What does

$$a^* \in B(s)$$

mean?

(b) What does

$$V(s) = \max_{a \in B(s)} u(a)$$

mean?

(c) What inequality must hold between $u(a^*)$ and $u(a)$ for an arbitrary $a \in B(s)$?

(d) Rewrite

$$a^* \in \text{argmax}_{a \in B(s)} u(a)$$

in words.

11 Step 5: Recognize When the Definition Does Almost All the Work

Not every proof requires a clever manipulation.

Consider:

Definition 1 (Homogeneous Function)

A function $f : \mathbb{R}^n_+ \to \mathbb{R}$ is homogeneous of degree k if

$$f(tx) = t^k f(x)$$

for every x in its domain and every $t > 0$.

Now suppose a problem asks:

> Suppose f is homogeneous of degree one. Show that
>
> $$f(tx) = tf(x) \qquad \forall t > 0.$$

There is essentially nothing to calculate.

The desired conclusion follows directly from the definition with $k = 1$.

The broader lesson is:

> Before searching for a theorem or calculation, ask whether the conclusion is already contained in a definition.

12 Exercise: Identify the Hidden Definition

For each statement, identify the definition that contains most of the information needed for the proof.

(a) Suppose f is homogeneous of degree zero. Show that

$$f(tx) = f(x) \qquad \forall t > 0.$$

(b) Suppose u is strictly concave and $x \neq y$. Show that

$$u\left(\frac{x+y}{2}\right) > \frac{u(x)+u(y)}{2}.$$

(c) Suppose A is negative definite. What can you conclude about

$$x^\top Ax$$

when $x \neq 0$?

(d) Suppose

$$x^* \in \mathrm{argmax}_{x\in C} f(x).$$

What comparison must hold between $f(x^*)$ and $f(x)$ for $x \in C$?

13 Writing a Proof Skeleton

Once a problem has been deconstructed, it is often useful to write a **proof skeleton** before filling in the details.

Return to the strict concavity result:

Let C be convex and let $f : C \to \mathbb{R}$ be strictly concave. Show that f has at most one maximizer.

A proof skeleton could look like:

1. Suppose there are two distinct maximizers x^* and y^*.

2. Since C is convex,

$$\underline{\hspace{6cm}}.$$

3. Since f is strictly concave,

$$\underline{\hspace{6cm}}.$$

4. Since x^* and y^* are maximizers,

$$\underline{\hspace{6cm}}.$$

5. Combining these statements gives

$$\underline{\hspace{6cm}},$$

which contradicts the definition of a maximizer.

The advantage of this approach is that it separates two tasks:

$$\boxed{\text{Logical structure} \qquad \text{from} \qquad \text{algebraic details.}}$$

14 Exercise: Write Only the Beginning

For each statement below, **do not finish the proof**.

Instead:

1. identify what is given;

2. rewrite what must be shown;

3. identify the relevant definition;
4. write a reasonable first one or two lines.

14.1 Increasing Functions

Suppose $f : \mathbb{R} \to \mathbb{R}$ and $g : \mathbb{R} \to \mathbb{R}$ are increasing. Show that $f + g$ is increasing.

14.2 Intersection of Convex Sets

Suppose $C_1, \ldots, C_K \subseteq \mathbb{R}^n$ are convex. Show that

$$\bigcap_{k=1}^{K} C_k$$

is convex.

14.3 Positive Definiteness

Suppose A is positive definite. Show that

$$x^\top A x = 0 \quad \Longrightarrow \quad x = 0.$$

14.4 Multiple Maximizers under Concavity

Let $C \subseteq \mathbb{R}^n$ be convex and let $f : C \to \mathbb{R}$ be concave.

Suppose

$$x^*, y^* \in \operatorname{argmax}_{x \in C} f(x).$$

Show that for every $\lambda \in [0, 1]$,

$$\lambda x^* + (1 - \lambda) y^*$$

is also a maximizer.

15 A More Economic Example

Consider the following consumer problem:

$$\max_{x \in \mathbb{R}^L_+} u(x) \qquad \text{subject to} \qquad p \cdot x \leq w.$$

Define the budget set

$$B(p, w) = \left\{ x \in \mathbb{R}^L_+ : p \cdot x \leq w \right\}.$$

Suppose u is strictly concave and $B(p, w)$ is convex.

A typical microeconomics statement is:

Proposition 1

If a utility-maximizing bundle exists, then it is unique.

Before writing a proof, identify the structure.

(a) What is the feasible set?

(b) What assumption tells you that convex combinations of feasible bundles remain feasible?

(c) What does it mean for x^* and y^* to both solve the consumer's problem?

(d) What inequality does strict concavity give you if $x^* \neq y^*$?

(e) How is this problem mathematically related to the earlier uniqueness proposition?

The economic vocabulary has changed, but the mathematical structure has not.

16 Final Workshop

For the final problem, do not begin by writing a proof.

Spend the first several minutes completing the deconstruction.

Proposition 1

Let $C \subseteq \mathbb{R}^n$ be convex. Suppose $f, g : C \to \mathbb{R}$ are convex and $\alpha, \beta \geq 0$.

Define

$$h(x) = \alpha f(x) + \beta g(x).$$

Show that h is convex.

1. What is given?

2. What exactly must be shown?

3. What definition should be unpacked?

4. What arbitrary objects should appear in the first line?

5. What does convexity of f give you?

6. What does convexity of g give you?

7. Why might the assumptions $\alpha, \beta \geq 0$ matter?

8. Write a proof skeleton before writing the proof.

"What trick am I supposed to use?"

Instead, begin with:

1. **Given:** What assumptions am I allowed to use?
2. **Goal:** What exactly must be established?
3. **Definitions:** What do the important words and symbols mean?
4. **Objects:** What arbitrary objects should I introduce?
5. **Assumptions:** What does each assumption allow me to conclude?
6. **Skeleton:** What intermediate steps would connect the assumptions to the goal?

$\boxed{\text{Understand first. Prove second.}}$